

ASSEMENT TOOL -1

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Q1. Security Attacks (10 Marks)

Analysis of Security Incidents

Incident	Type of Attack	CIA Impact	Preventive Measures
Customers received fake emails requesting login credentials	Active Attack (Phishing/Social Engineering)	Confidentiality: Login credentials stolen. Integrity: Unauthorized transactions possible. Availability: Indirect impact if accounts are blocked.	Email filtering, Multi-Factor Authentication (MFA), employee/customer awareness training, anti-phishing solutions.
Unauthorized fund transfers using intercepted session tokens	Active Attack (Session Hijacking)	Confidentiality: Session information exposed. Integrity: Unauthorized fund transfers alter banking records. Availability: Customers may lose account access.	HTTPS/TLS encryption, secure session management, session timeout, secure cookies, token regeneration.
Banking server experienced repeated traffic spikes	Active Attack (Distributed Denial-of-Service – DDoS)	Confidentiality: No direct impact. Integrity: Normally unaffected. Availability: Banking services become unavailable.	Firewalls, DDoS mitigation services, Intrusion Prevention Systems (IPS), traffic filtering, load balancing.
Employee	Active Attack	Confidentiality:	Strong password



accounts compromised through weak passwords	(Password Attack/Brute Force)	Sensitive data accessed. Integrity: Data can be modified. Availability: Accounts may be locked or misused.	policy, MFA, password managers, account lockout policies, security awareness training.
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Conclusion

The banking system faces multiple active attacks affecting the CIA triad. Implementing strong authentication, encryption, firewalls, and user awareness programs significantly improves security.

Q2. Security Mechanisms and Security Services (10 Marks)

Security Requirements and Solutions

Requirement	Security Service	Security Mechanism	Justification
Only authorized doctors access records	Authentication, Access Control	MFA, Role-Based Access Control (RBAC), Biometric Authentication	Ensures only authorized users can access patient records.
Patient records cannot be modified	Integrity	Hashing (SHA-256), Digital Signatures	Detects unauthorized modifications.
Every medical transaction must be traceable	Non-repudiation	Digital Signatures, Audit Logs	Prevents users from denying performed actions.
Secure communication between hospitals	Confidentiality	AES Encryption, TLS/SSL, VPN	Protects patient information during transmission.
System continues during cyber attacks	Availability	Firewalls, IDS/IPS, DDoS Protection, Backup Servers	Maintains uninterrupted healthcare services.

Recommended Security Mechanisms

- AES-256 Encryption



- TLS/SSL
- Digital Signatures
- SHA-256 Hashing
- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)
- Firewalls
- Intrusion Detection System (IDS)
- Intrusion Prevention System (IPS)
- Regular Backups

Conclusion

Combining security services with modern security mechanisms ensures confidentiality, integrity, availability, authentication, access control, and non-repudiation within the EHR system.

Q3. Caesar Cipher (10 Marks)

Given Ciphertext

YMNX NX F XJHWJY RJXXFLJ

Step 1: Determine the Key

Using brute-force analysis, Encryption Key = 5

Step 2: Decrypt the Message

Shift every letter 5 positions backward.

Plaintext: THIS IS A SECRET MESSAGE

Step 3: Why Caesar Cipher is Vulnerable

- Only 25 possible keys
- Easily broken using brute-force attack



- Vulnerable to frequency analysis
- Provides very low security
- Not suitable for modern communication

Step 4: Recommended Encryption

Advanced Encryption Standard (AES-256)

Justification

- 256-bit key
- Extremely resistant to brute-force attacks
- Fast and efficient
- Industry standard
- Used in banking, military, and government systems

Conclusion

Caesar Cipher is obsolete and should be replaced with AES-256 for secure logistics communication.

Q4. Playfair Cipher (10 Marks)

Keyword: DEFENSE

Step 1: Construct Playfair Matrix (Combine I/J)

D	E	F	N	S
A	B	C	G	H
I/J	K	L	M	O
P	Q	R	T	U
V	W	X	Y	Z

Step 2: Ciphertext and Decryption

Ciphertext: BMODZBXDNABEKUDMUIXMMOUIVIF

Decrypted Plaintext: HIDETHEGOLDINTHETREESTUMP



Step 3: Advantages over Caesar Cipher

- Encrypts letter pairs instead of single letters.
- Reduces frequency analysis effectiveness.
- Larger key space.
- More resistant to simple attacks.
- Better confidentiality.

Step 4: Limitations

- Still vulnerable to cryptanalysis.
- Manual encryption is slow.
- Key distribution is difficult.
- Not secure against modern computing attacks.
- Cannot protect military communication against advanced cyber threats.

Recommendation

Use AES-256 together with RSA for secure key exchange.

Conclusion

Playfair Cipher provides better security than Caesar Cipher but is inadequate for protecting modern military communications.

Q5. Integrated Case Study – Security Analysis (10 Marks)

Vulnerability Analysis

Issue	Security Attack
Caesar Cipher used for machine messages	Weak Encryption
Playfair Cipher for supplier communication	Weak Classical Encryption
Employees receive phishing emails	Phishing Attack
Production schedules modified	Unauthorized Access / Insider Attack
Production server under attack	Distributed Denial-of-Service (DDoS)



Required Security Services

Issue	Security Service
Weak encryption	Confidentiality
Phishing	Authentication
Unauthorized modification	Integrity
DDoS	Availability
Supplier communication	Confidentiality, Authentication
Employee access	Access Control
Audit tracking	Non-repudiation

Recommended Security Mechanisms

Problem	Solution
Caesar Cipher	AES-256 Encryption
Playfair Cipher	AES + RSA
Phishing	MFA, Email Filtering, Security Awareness Training
Unauthorized Access	RBAC, Digital Signatures, Hashing
DDoS	Firewalls, IDS/IPS, DDoS Protection
Authentication	Biometrics, MFA
Monitoring	SIEM, Continuous Log Monitoring

Overall Conclusion

The organization's cybersecurity posture can be significantly strengthened by replacing outdated classical encryption methods with modern cryptographic algorithms such as AES-256 and RSA. Implementing Multi-Factor Authentication, Role-Based Access Control, Digital Signatures, Firewalls, IDS/IPS, DDoS protection, and continuous monitoring will ensure confidentiality, integrity, availability, authentication, access control, and non-repudiation.

